

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 6

10 Semantics of Programming Languages (nk480)

Consider the language with functions, integers, and printing.

$$\begin{array}{ll} \tau ::= \text{unit} \mid \text{int} \mid \tau \rightarrow \tau' & \text{Types} \\ e ::= x \mid \lambda x : \tau. e \mid e e' \mid \text{skip} \mid n \mid \text{print}(e) \mid e; e' & \text{Terms} \end{array}$$

The typing rule for $\text{print}(e)$ is:

$$\frac{\Gamma \vdash e : \text{int}}{\Gamma \vdash \text{print}(e) : \text{unit}}$$

- (a) Define a small-step, call-by-value operational semantics for this language. Clearly explain what the components of the machine configuration are, and how it identifies what is printed. [10 marks]

Answer: First, we define the values and evaluation contexts:

$$\begin{array}{ll} v ::= \text{skip} \mid n \mid \lambda x : \tau. e \\ E ::= \square e \mid v \square \mid \text{print}(\square) \mid \square; e \end{array}$$

Now, we can use this to define a small-step semantics on a machine configuration $\langle e; \omega \rangle$, where ω is a list of the numbers which have been printed so far. Note that we use a list to keep track of the order of the printed numbers.

$$\frac{\langle e; \omega \rangle \rightsquigarrow \langle e'; \omega' \rangle}{\langle E[e]; \omega \rangle \rightsquigarrow \langle E[e']; \omega' \rangle}$$

$$\overline{\langle (\lambda x : \tau. e) v; \omega \rangle} \rightsquigarrow \overline{\langle [v/x]e; \omega \rangle}$$

$$\overline{\langle \text{skip}; e; \omega \rangle} \rightsquigarrow \overline{\langle e; \omega \rangle}$$

$$\overline{\langle \text{print}(n); \omega \rangle} \rightsquigarrow \overline{\langle \text{skip}; \omega \cdot n \rangle}$$

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- (b) State a progress theorem for this language, and explain what it says about the evolution of the machine state. [4 marks]

Answer: For all e, τ , and ω , if $\cdot \vdash e : \tau$ then e is a value or $\langle e; \omega \rangle \rightsquigarrow \langle e'; \omega' \rangle$ for some e' and ω' such that ω is a prefix of ω' .

This progress property ensures that printed values are never forgotten. It is possible to prove a weaker theorem omitting the prefix property, but that is not worth full marks, since all it will say is that there is *some* new output at each step.

- (c) Prove progress for the $\text{print}(e)$ case, giving the names of any of the standard properties (such as substitution) that you needed to use in the proof. [6 marks]

Answer: The typing rule for `print(e)` is:

$$\frac{\Gamma \vdash e : \text{int}}{\Gamma \vdash \text{print}(e) : \text{unit}}$$

Since the proof is by rule induction on the typing derivation, we know that either e is a value or $\langle e; \omega \rangle \rightsquigarrow \langle e'; \omega' \rangle$ for some e' and ω' such that ω is a prefix of ω' .

- Case e is a value:

By inversion, we know e is some numeral n .

By reduction, we know that $\langle \text{print}(n); \omega \rangle \rightsquigarrow \langle \text{skip}; \omega \cdot n \rangle$. We take $\omega \cdot n$ as the witness for ω' , and note that ω is trivially a prefix of ω .

- Case $\langle e; \omega \rangle \rightsquigarrow \langle e'; \omega' \rangle$:

By a congruence rule in reduction, we know that $\langle \text{print}(e); \omega \rangle \rightsquigarrow \langle \text{print}(e'); \omega' \rangle$.

Induction also told us ω is a prefix of ω' .
